Fourier Series and PDEs Problem Sheet 3

1. Let f be a continuous periodic function of period $2L$. For positive integers N, denote by

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right)$$

the partial sums of the Fourier series for f, and let T_N be the sum defined by

$$T_N(x) = \frac{A_0}{2} + \sum_{n=1}^{N} \left(A_n \cos\left(\frac{n\pi x}{L}\right) + B_n \sin\left(\frac{n\pi x}{L}\right)\right),$$

where A_n and B_n are arbitrary real constants.

(a) State integral expressions over $[-L, L]$ for the Fourier coefficients a_n and b_n.

(b) Show that

$$\frac{1}{L}\int_{-L}^{L} f(x)T_N(x)\,\mathrm{d}x = \frac{a_0A_0}{2} + \sum_{n=1}^{N} \big(a_nA_n + b_nB_n\big).$$

Hence write down similar expressions for

$$\frac{1}{L}\int_{-L}^{L} T_N(x)^2\,\mathrm{d}x, \qquad \frac{1}{L}\int_{-L}^{L} f(x)S_N(x)\,\mathrm{d}x \qquad \text{and} \qquad \frac{1}{L}\int_{-L}^{L} S_N(x)^2\,\mathrm{d}x.$$

[*You may find it helpful to note that (i) if $f = T_N$ then $a_n = A_n$ and $b_n = B_n$ for $n \leq N$ and (ii) if $T_N = S_N$ then $A_n = a_n$ and $B_n = b_n$ for $n \leq N$.*]

(c) Deduce that

$$\mathcal{E}(T_N) - \mathcal{E}(S_N) = \frac{(A_0 - a_0)^2}{2} + \sum_{n=1}^{N} \Big((A_n - a_n)^2 + (B_n - b_n)^2\Big),$$

where the *mean-squared error* in the approximation of f by T_N is defined by

$$\mathcal{E}(T_N) = \frac{1}{L}\int_{-L}^{L} \big(T_N(x) - f(x)\big)^2\,\mathrm{d}x.$$

Hence write down the values of the constants A_n and B_n for which $\mathcal{E}(T_N)$ is minimized for each positive integer N.

2. The periodic function f of period $2L$ is defined by

$$f(x) = \begin{cases} 1 + x/L & \text{for } -L < x \leqslant 0, \\ 0 & \text{for } 0 < x \leqslant L. \end{cases}$$

(a) Sketch the graph of $f(x)$ for $x \in (-2L, 2L]$, indicating any point x at which the Fourier series for f does not converge to $f(x)$ and showing the values to which the Fourier series does converge at these points.

(b) Show that the Fourier series for f is given by

$$f(x) \sim \frac{1}{4} + \sum_{m=1}^{\infty} \left(\frac{2}{(2m-1)^2\pi^2} \cos\left(\frac{(2m-1)\pi x}{L}\right) - \frac{1}{m\pi} \sin\left(\frac{m\pi x}{L}\right)\right).$$

(c) By considering the value of the series at $x = L/2$ and at $x = 0$, deduce that

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4} \qquad \text{and} \qquad \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

3. (a) The function f is defined by

$$f(x) = \exp(x/L) \quad \text{for } 0 \leq x \leq L.$$

(i) Evaluate the Fourier cosine and sine series for f on $[0, L]$.

(ii) For each series sketch the graph of the function to which it converges on $(-2L, 2L]$.

(iii) Describe three ways in which the truncated cosine series for f is a better approximation of f on $[0, L]$ than the truncated sine series for f.

(b) Give, with justification, an example of a polynomial that is better approximated on $[0, L]$ by its truncated sine series than its truncated cosine series.

4. The function f is continuous on $[0, L]$ and has piecewise continuous derivative on $(0, L)$, where L is a positive constant.

(a) Suppose that f may be expressed in the form of the generalized Fourier series

$$f(x) = \sum_{n=0}^{\infty} c_n \sin\left(\frac{(2n+1)\pi x}{2L}\right) \quad \text{for} \quad 0 < x < L.$$

Assuming that the orders of summation and integration may be interchanged, derive integral expressions over $[0, L]$ for the Fourier coefficients c_n.

[*You may assume without proof the orthogonality relations*

$$\int_0^L \sin\left(\frac{(2m+1)\pi x}{2L}\right) \sin\left(\frac{(2n+1)\pi x}{2L}\right) \mathrm{d}x = \frac{L}{2}\delta_{mn},$$

where m and n are non-negative integers and δ_{mn} is Kronecker's delta.]

(b) To which function does the series in part (a) converge for $x \in \mathbb{R}$? Justify your answer by applying the *Fourier Convergence Theorem* to a suitable $4L$-periodic extension for f.